Build a Human Classifier

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

Recall and choose your starting point

> Record your starting explanation for Build a Human Classifier before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Learning from Examples

Without opening the old lesson, explain one key idea from Learning from Examples. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Repeating Efficiently

Without opening the old lesson, explain one key idea from Repeating Efficiently. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Features: Useful Clues

Without opening the old lesson, explain one key idea from Features: Useful Clues. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Build a Human Classifier trace and record the first step you can explain without help.

Use W2: Construct both outcome paths for Constructing a **classifier**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Build a Human Classifier explanation against the supplied facts before discussing it.

Use the worked case for Constructing a **classifier**. Proposed explanation: “Any numeric **input** must receive a category”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Build an independent ribbon classifier

Design Ribbon1 for ribbon lengths recorded as whole centimetres from 1 through 12 inclusive, with unit cm required. Predict long at 8 cm or more and short otherwise.

Write complete instructions and trace 7 cm, 8 cm, a blank length and 13 cm. Include reasons for **no prediction** and the **model version** in each record.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Build a Human Classifier target to a trace and a justified conclusion.

A classroom rule labels anything with wings as a bird. Test a bat, a penguin and a toy aeroplane. Define the target independently, then propose a next step without pretending one visual clue solves biology.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Build a Human Classifier case and show the deciding evidence.

A human **classifier** uses has wheels means vehicle on a bicycle, wheelchair, toy car and boat.

Apply the rule and identify a limitation.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Use a human **classifier**: three straight sides forming a closed shape means triangle; four equal straight sides and four right angles forming a closed shape means square. Classify a three-sided card and a card whose corner is hidden. When should you pause?

My independent answer and reason

Write or draw your reasoning and evidence.

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Build a Human Classifier adds to Helpful Classroom Sorter and keep the evidence.

Make the sorter's current **decision boundary** visible as a rule chart or feature map.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A classroom rule labels anything with wings as a bird. Test a bat, a penguin and a toy aeroplane. Define the target independently, then propose a next step without pretending one visual clue solves biology.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

W1 Write an input contract

> Write an input contract. Apply the supplied Build a Human Classifier facts and keep your reasoning.

Write the complete Tower1 **input contract**. Give one accepted example and three different rejected examples, each with its reason. Explain why an accepted **input** does not have to be one of the examples used to fit the threshold.

My reasoning and evidence

W2 Construct both outcome paths

> Construct both outcome paths. Apply the supplied Build a Human Classifier facts and keep your reasoning.

Write numbered instructions for Tower1, including **input** checks, the stopping path after a failed check, the threshold rule and the record to keep. Your partner must be able to use them without supplying an unstated rule.

> **My reasoning and evidence**

W3 Trace boundaries and a broken path

> Trace boundaries and a broken path. Apply the supplied Build a Human Classifier facts and keep your reasoning.

Trace 1 cm, 5 cm, 6 cm, 10 cm and 11 cm using Tower1. A faulty procedure records **no prediction** for a blank, then continues and treats the blank as zero to report short.

Identify the missing instruction and the proper final outcome.

> **My reasoning and evidence**

WORKBOOK · 15 OF 21

W4 Build an independent ribbon classifier

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Build an independent ribbon classifier. Apply the supplied Build a Human Classifier facts and keep your reasoning.

Design Ribbon1 for ribbon lengths recorded as whole centimetres from 1 through 12 inclusive, with unit cm required. Predict long at 8 cm or more and short otherwise.

Write complete instructions and trace 7 cm, 8 cm, a blank length and 13 cm. Include reasons for **no prediction** and the **model version** in each record.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Explain what your procedure establishes

> Explain what your procedure establishes. Apply the supplied Build a Human Classifier facts and keep your reasoning.

Ribbon1 accepts 12 cm and predicts long. No comparison label has been supplied for that record. State what has been checked and what remains unknown.

Explain why changing a blank to 1 cm just to produce a category would be a mistake.

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Build a Human Classifier

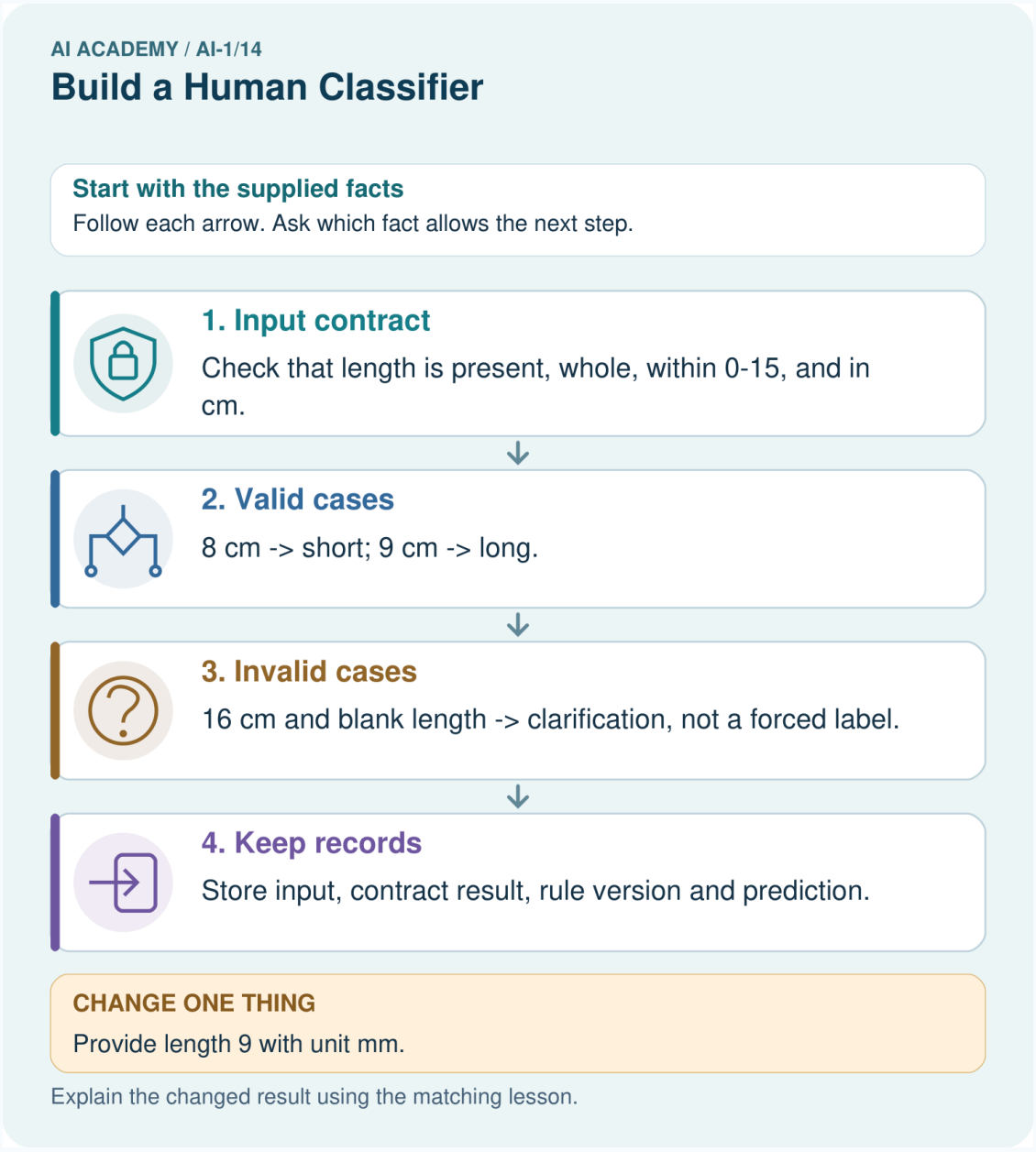
> A fictional ribbon sorter uses a fixed instruction card called RibbonR. Explain the labelled step

> before checking the changed condition.

A fictional ribbon sorter uses a fixed instruction card called RibbonR.

Valid **input**: whole-number length from 0 to 15 with unit cm.

Predict long at least 9 cm; otherwise short. Missing, wrong-unit or out-of-range values request clarification.



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace 0 cm and explain why clarification is different from predicting short.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Create a sorter instruction card covering valid values, equality, missing fields and who can resolve clarification requests.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Helpful Classroom Sorter record.

Project: Helpful Classroom Sorter

Current milestone: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Create a sorter instruction card covering valid values, equality, missing fields and who can resolve clarification requests.

Evidence to keep	My record
This week's input and deciding step	

Evidence to keep	My record
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional ribbon sorter uses a fixed instruction card called RibbonR.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Store **input**, contract result, rule version and **prediction**.

1. Invalid cases

2. **Input contract**

3. Keep records

4. Valid cases

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Keep records Store **input**, contract result, rule version and **prediction**. A **prediction** boundary answers a different question from **input validation**.

Length 9 passes the contract and meets the long boundary. Length 16 must not receive long merely because it exceeds 9: it violates the allowed range first. Zero is permitted and receives short.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Provide length 9 with unit mm. Which conclusion is supported by the supplied facts?

1. Request clarification rather than predicting long.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Request clarification rather than predicting long. RibbonR requires cm; the unit mismatch fails its declared **input contract**.

Repair a missing building block

Review Learning from Examples (ai-1/13). Goal: Fit a threshold by tracing **prediction**s and counting disagreements for every candidate. Short practice: Which new measured length would reveal disagreement between the candidates?.

Return to this week and explain what you changed.

Review Repeating Efficiently (ai-1/11). Goal: Trace a loop by testing its condition before every repetition. Short practice: Trace a fresh start of 9. How many additions occur, and must the report equal 10?.

Return to this week and explain what you changed.

Review Features: Useful Clues (ai-1/8). Goal: Choose recorded properties that preserve distinctions needed for a stated task.

Short practice: Point to the pair that tests pattern, then explain why the sticker is a bad early **input**.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

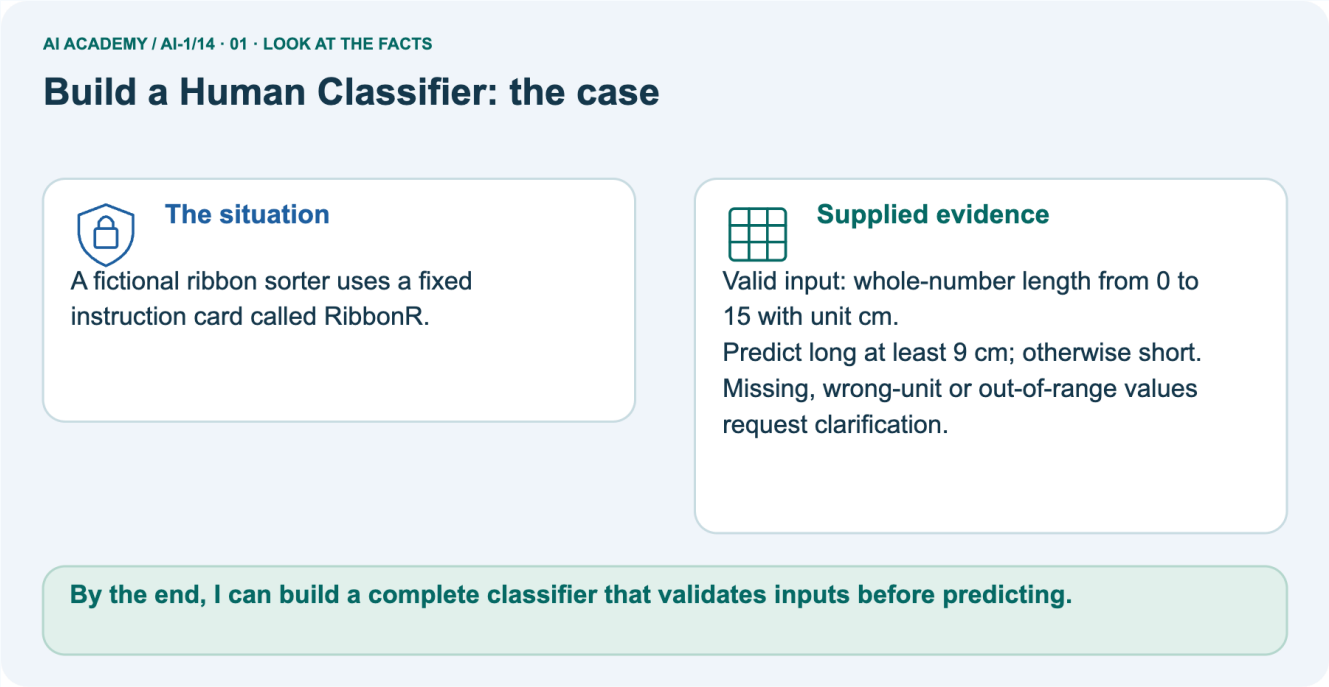
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Build a Human Classifier picture from its labels, then record a relevant limit.

01. Build a Human Classifier: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

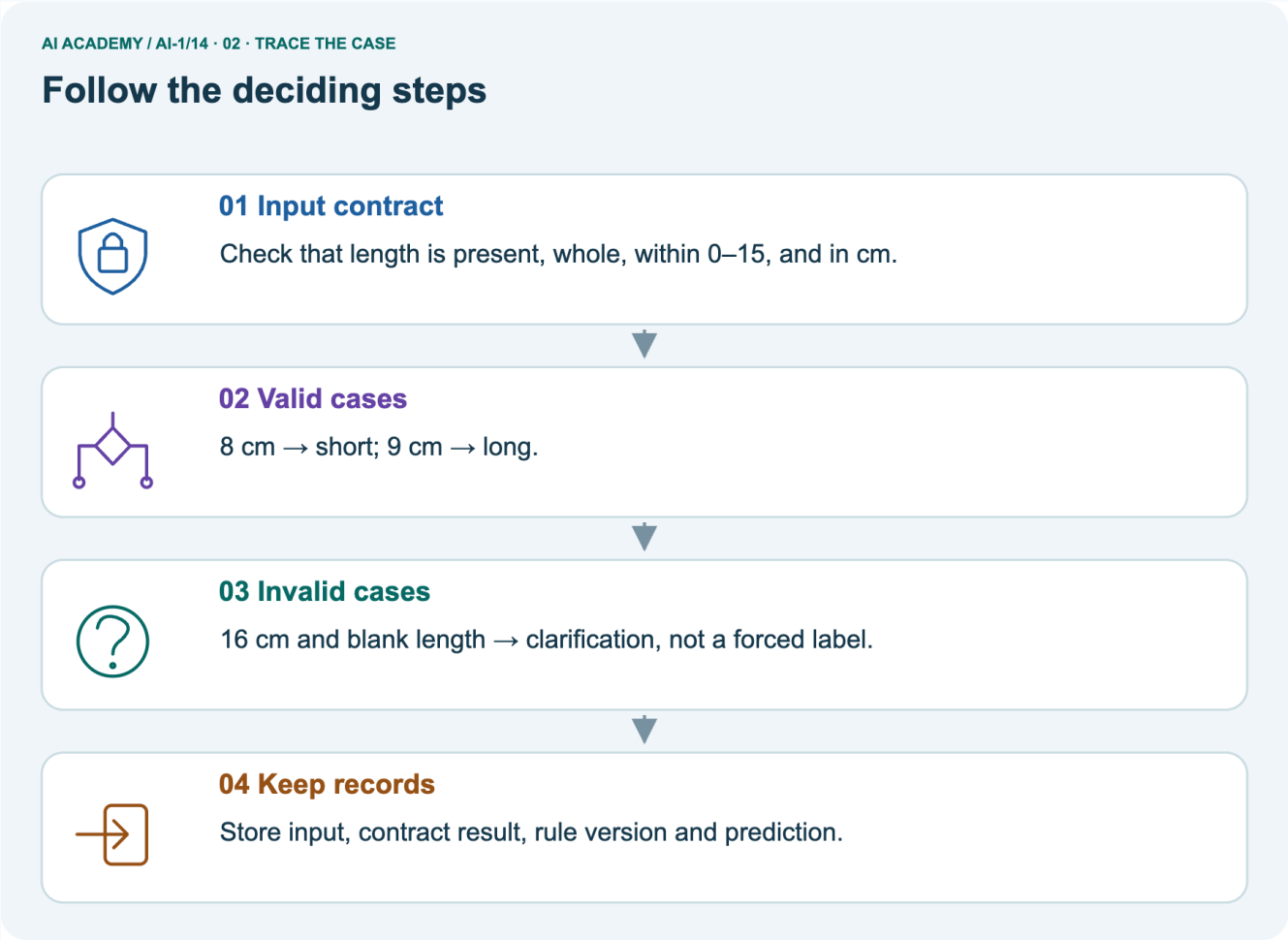
A fictional ribbon sorter uses a fixed instruction card called RibbonR. Supplied facts: Valid **input**: whole-number length from 0 to 15 with unit cm. Predict long at least 9 cm; otherwise short.

Missing, wrong-unit or out-of-range values request clarification.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



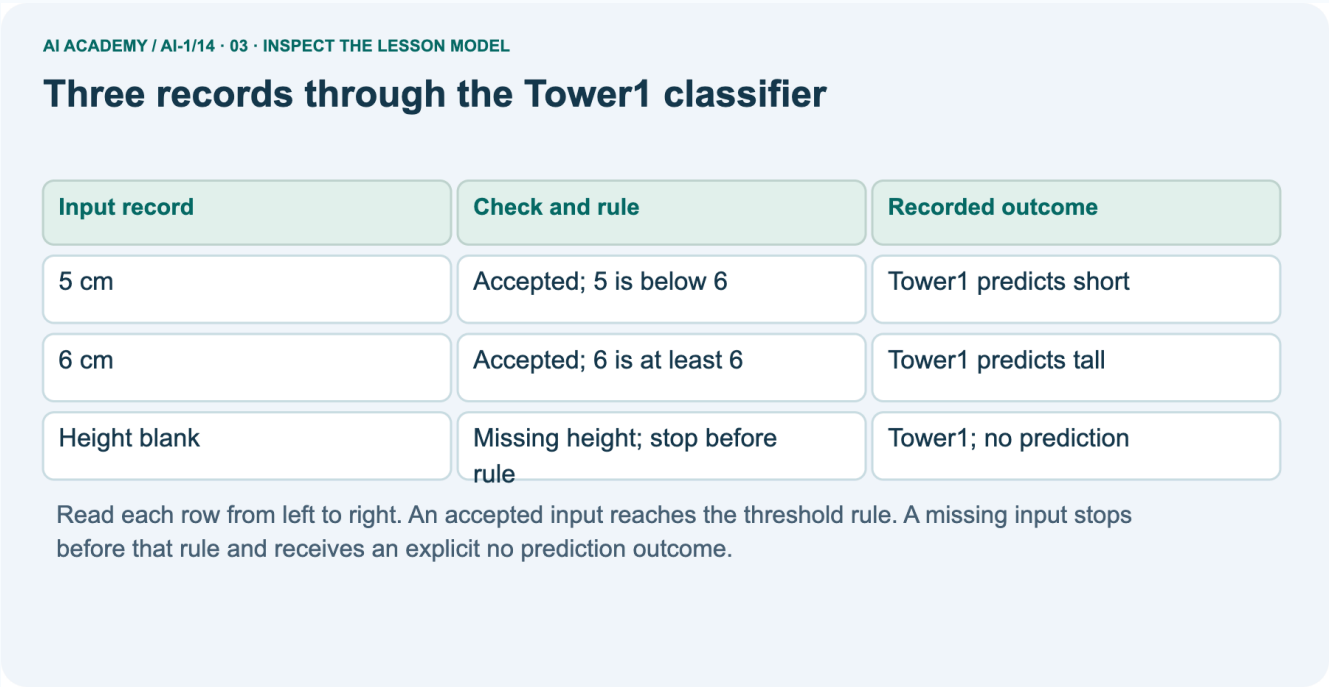
Follow the labelled picture with the matching explanation. Open the vector diagram

Input contract: Check that length is present, whole, within 0–15, and in cm. → Valid cases: 8 cm → short; 9 cm → long. → Invalid cases: 16 cm and blank length → clarification, not a forced label. → Keep records: Store **input**, contract result, rule version and **prediction**.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Three records through the Tower1 classifier



AI ACADEMY / AI-1/14 · 03 · INSPECT THE LESSON MODEL

Three records through the Tower1 classifier

Input record	Check and rule	Recorded outcome
5 cm	Accepted; 5 is below 6	Tower1 predicts short
6 cm	Accepted; 6 is at least 6	Tower1 predicts tall
Height blank	Missing height; stop before rule	Tower1; no prediction

Read each row from left to right. An accepted input reaches the threshold rule. A missing input stops before that rule and receives an explicit no prediction outcome.

Follow the labelled picture with the matching explanation. Open the vector diagram

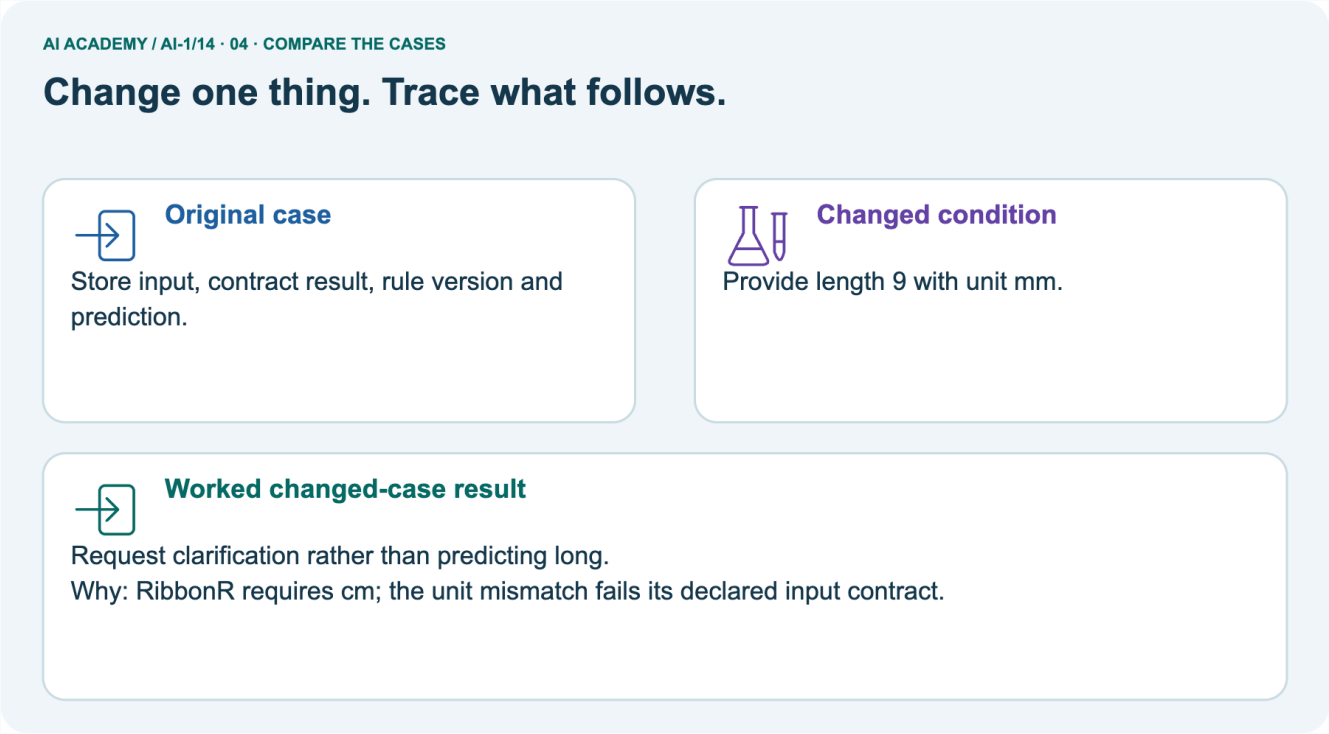
Three records through the Tower1 **classifier**. 5 cm | Accepted; 5 is below 6 | Tower1 predicts short. 6 cm | Accepted; 6 is at least 6 | Tower1 predicts tall.

Height blank | Missing height; stop before rule | Tower1; **no prediction**

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

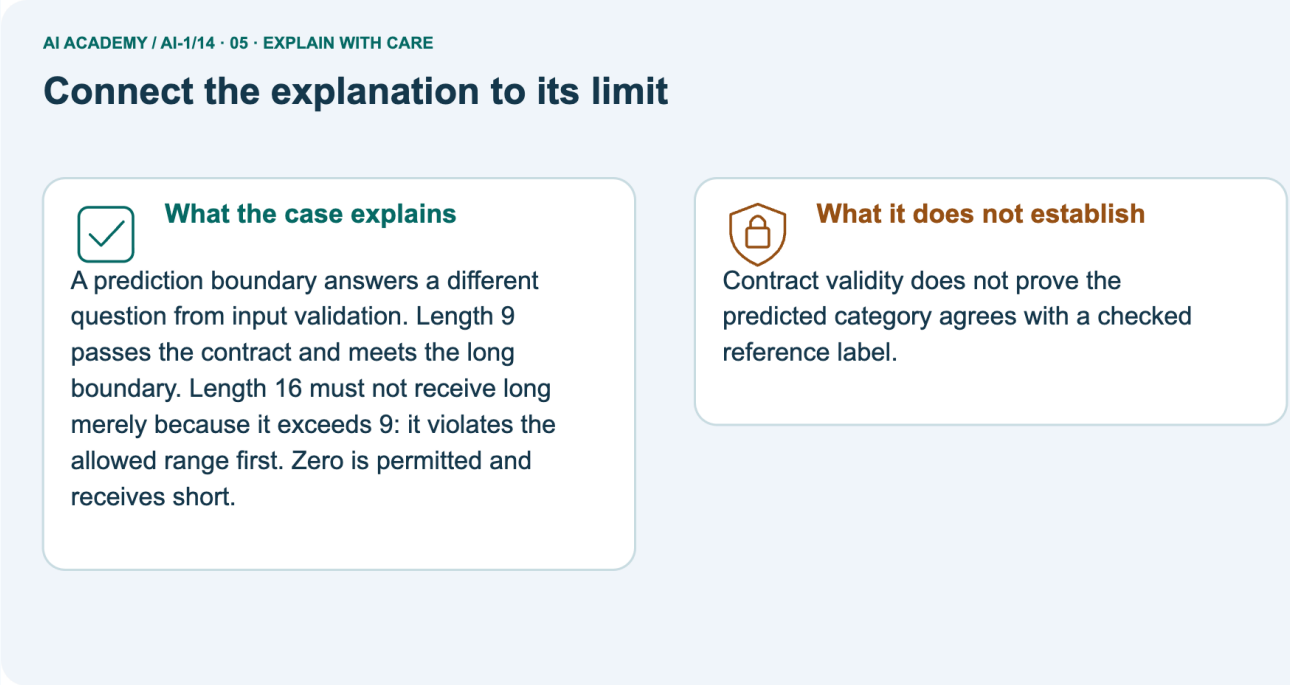
Original case: Store **input**, contract result, rule version and **prediction**. Changed condition: Provide length 9 with unit mm. Expected result: Request clarification rather than predicting long.

Reason: RibbonR requires cm; the unit mismatch fails its declared **input contract**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **prediction** boundary answers a different question from **input validation**. Length 9 passes the contract and meets the long boundary.

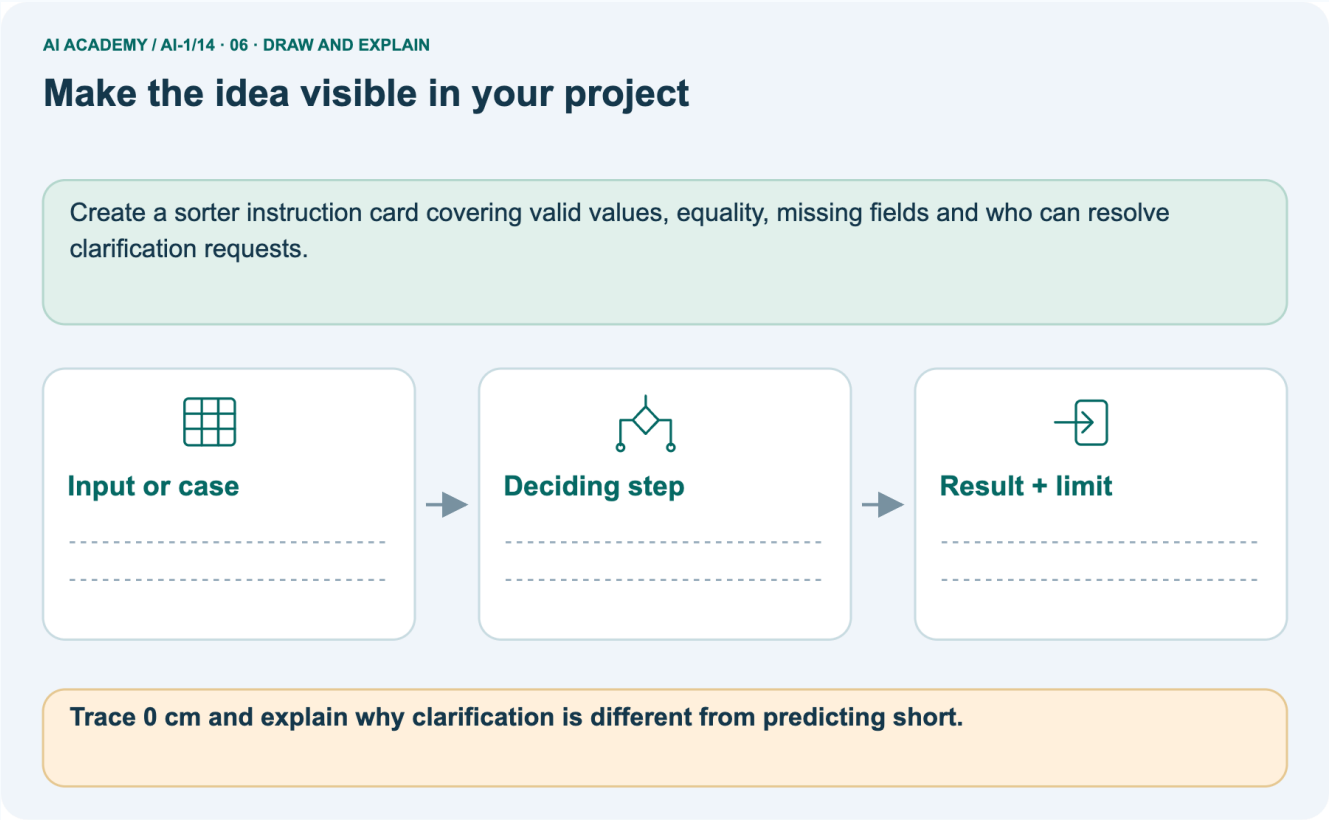
Length 16 must not receive long merely because it exceeds 9: it violates the allowed range first. Zero is permitted and receives short.

Limit: Contract validity does not prove the predicted category agrees with a checked reference label.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Create a sorter instruction card covering valid values, equality, missing fields and who can resolve clarification requests. Draw your **input**, deciding step, result and limit.

Supported question: Trace 0 cm and explain why clarification is different from predicting short.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.